

Statistical Analysis Assignment

Energy mix and CO₂ Emissions

Degree in Management Engineering | University of Ruse Erasmus | Statistics





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1. OBJECTIVES

The objective of this assignment is to analyze the relationship between the energy mix (coal, natural gas, renewable energy, nuclear energy) and the CO₂ emissions across different countries, with a special focus on European nations.

2. RESEARCH QUESTIONS

- How do different energy sources affect CO₂ emissions per capita?
- Is there a **significant relationship** (correlation) between renewable energy share and emissions?
- Can countries be grouped based on their energy profiles and emission levels?
- Is there correlation or influence between the factors of energy sources and the economy status of the countries?

3. DATA COLLECTION AND CLEANING

For this part of the project, I will be using excel, with the Analysis ToolPak add-in. I will use the data from “OurWorldInData” about Global Energy Substitution, Life expectancy, Carbon Oxides Emissions per capita and GDP per capita.

The assignment is based on testing the relationship between renewable **energy share** and emissions. Because the input data breaks renewables down into four separate columns (Biofuels, Hydro, Solar, Wind), we will use Excel to add them up.

To start with, I selected 8 European Union countries, plus the US and China for a real comparison. I will only analyze data from 1980 to 2024, since it is most representative period for renewables.

After selecting countries and year period, I imported all the other data with the XLOOKUP formula. I managed to pull the data of CO₂ per capita, GDP per capita, Life Expectancy, and Energy. Then I added them all up in the column for Total renewables.

Since we need to match the country and the year both at the same time, we will use a special trick: the & symbol. This symbol glues the Country and Year together, so Excel will look for "Bulgaria2000" for example.

For calculating the average, stdev and variance of the GDP per capita and CO2, I used pivot tables in excel. I believe that for this continuous data they are the best measures. I also decided to calculate the COV separately, as it could be a great indicator too. I used 3 representative figures after the decimal point for all calculations.

Then comes the calculation for the growth rate, which will return how much these values have grown in 20 years.

My choice of years was selective, as I made it only between 2000 and 2020 because data for renewable energy before the 2000's is not as representative, and because I wanted to keep a 20-year timeframe.

I made the calculations by using pivot tables once again. A negative growth rate means that the country reduced its CO2 emissions.

Then comes constructing the confidence interval of the mean of the factors by years (taking the means of all countries and calculating the confidence interval for each year). Once again, I used pivot tables.

We need three pieces of information for every single year to build a confidence interval: the **Mean**, the **Standard Deviation**, and the **Count** of countries we are measuring. After computing the average, we add and subtract it to the margin of error (computed with the formula =CONFIDENCE) with a confidence interval of 95%. With this we have obtained 95% confidence intervals.

For calculating the T-tests, let's test if there is a statistically significant difference in CO2 emissions between **France** and **Germany** from 2000 to 2020.

I selected 20 years of data, filtering for France and Germany (2000-2020). In the data tab we scroll to the t-test: two sample assuming unequal variances (variable 1-France, variable 2-Germany). Checking the box "labels" and setting the output range on an empty cell, we obtain the results of the test.

- If the $P(T \leq t)$ two-tail number is **less than 0.05**, you write in your report:
- If it is higher than 0.05, there is no significant difference.

Since our result is $3.01526E-19$, which is almost negligible, we can affirm that there is a statistically significant difference in CO2 emissions between France and Germany.

After that, I started testing hypothesis about significant difference between mean of countries and Europe. Excel's Toolpak doesn't have a 1-click button for a "One-Sample T-Test," so I calculated the p-value using a few quick formulas. Let's test if **Italy's** CO2 emissions are significantly different from the **Overall European Average**.

In the EDA_Stats pivot table we select the "Grand Total" row. The grand total average for CO2 is the population mean.

For Italy's stats, in the EDA_Stats pivot table we take in its row

- **Sample Mean:** Italy's Average CO2
- **Standard Deviation:** Italy's StdDev for CO2
- **Count:** 21 (since we are looking at 20 years)

To calculate the T-statistic and P-value, I used this formula:

$$=([Italy\ Mean]-[Europe\ Mean])/([Italy\ StdDev]/SQRT(21))$$

To calculate the P-value, I used this other formula:

$$=T.DIST.2T(ABS("t-statistic"), 20) \text{ (Note: The 20 is your degrees of freedom, which is } n-1, \text{ so } 21-1).$$

Since the p-value is also negligible (6.23564E-08), we can conclude that Italy's emissions are significantly different from the rest of Europe.

With this calculation we mark the step of EDA as finished. We can start with the correlation analysis.

4. CORRELATION ANALYSIS

The goal here is to see if factors are moving together. For example, as a country's Renewable Energy percentage goes up, does their CO2 per capita go down?

First step is to generate a **Correlation Matrix** for a specific country and then mathematically prove if that relationship is statistically significant.

To calculate the correlation coefficient, we use the Data Analysis tab once again and select “correlation”. For the input range, we highlight all the data of CO2, GDP and Total renewable % for 2000–2020 for France and Spain.

- A number close to **1** means a strong positive relationship (e.g., as GDP goes up, CO2 goes up).
- A number close to **-1** means a strong negative relationship (e.g., as Renewables go up, CO2 goes down).
- A number close to **0** means no relationship.

We now must check if these correlation numbers are statistically significant. To do this, we calculate the t-statistic and p-value for the relationship.

First is to calculate the t-statistic. Supposing the data from the correlation number is in the F3 box, we must use the formula:

$$=(F4*SQRT(21-2))/SQRT(1-(F4*F4))$$

Then we must calculate the p-value using the following formula:

$$=T.DIST.2T(ABS("t-statistic"),21-2)$$

The p-value being 6.69121E-05 means that there is a significant relationship between renewable energy adoption and CO2 emissions in France.

I repeated this same process for Spain and Denmark, and both returned similar results. Remember that a correlation number close to -1 means that, as renewable energy use grows, CO2 emissions go down.

Spain:

- correlation number = -0.859313699
- t-statistic = -7.324
- p-value = 6.06563E-07

Denmark:

- correlation number = -0.965254134
- t-statistic = -16.101
- p-value = 1.57313E-12

The case for Bulgaria was different.

Bulgaria:

- correlation number = -0.358792661
- t-statistic = -1.676
- p-value = 0.110215559

While there is a weak negative correlation coefficient of -0.358 , the high p-value indicates that we do not have enough statistical evidence to prove a reliable relationship between the increase of renewables and the decrease in CO₂ emissions per capita. This case is probably one of the odd ones out of Europe as it does not follow the trend of reducing heavily the emissions of CO₂.

On the other hand, this is the data for China:

China:

- correlation number: 0.859559758
- t-statistic = 7.332
- p-value = 5.97159E-07

Here we can see a big contrast with the average of European countries, as the strong positive correlation and highly significant p-value implies that their rapid economic expansion and overall energy demand completely outpaced their emissions-reduction efforts.

This aligns with the real-life data that China has been augmenting its total CO₂ emissions since 2000 and before, as the country is known for being a powerhouse of manufacturing and fast industrialization.

5. REGRESSION MODELING

Correlation just told us if two things move together. Multiple Regression will tell us exactly how much one factor influences CO2 emissions while controlling the other factors.

I built an excel model that predicts **CO2 Emissions** based on multiple independent variables, such as **GDP** and **Renewable Energy %**.

I will first show the build for the model of Bulgaria and then compare its results with the ones of Spain and Denmark.

- These are the results for **Bulgaria**:

Regression Statistics								
Multiple R	0.947775805							
R Square	0.898278977							
Adjusted R Square	0.880328209							
Standard Error	0.176117068							
Observations	21							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	3	4.65642201	1.5521407	50.04125	1.19638E-08			
Residual	17	0.527292769	0.0310172					
Total	20	5.183714779						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	1.196440821	0.485036397	2.4667032	0.024563	0.173103474	2.21977817	0.173103474	2.219778167
GDP_Per_Capita	2.06004E-05	2.39144E-05	0.8614222	0.400993	-2.98546E-05	7.1055E-05	-2.98546E-05	7.10554E-05
Total_Renewable_%	0.000122057	7.1858E-05	1.6985839	0.107624	-2.95503E-05	0.00027366	-2.95503E-05	0.000273664
Coal_%	0.000416218	3.48329E-05	11.949014	1.07E-09	0.000342727	0.00048971	0.000342727	0.000489709

The first thing we must put our eyes on is the R Square value. This value tells you how accurate your overall model is. It is a percentage between 0 and 1.

We can check that our R Square value is 0.898278977, which means that our model explains approximately 89% of the variance in Bulgaria's CO2 emissions. The remaining 11% is due to factors not included in this model.

Then we must look at the significance factor. This is the p-value for the entire model. Thereby, if this number is less than 0.05, the overall model is statistically significant.

Because our significant factor here is 1.19638E-08 (almost negligible), we can conclude that our study is indeed reliable.

The last part is analyzing each specific p-value. On the bottom table and just as before, if a factor's p-value is < 0.05, it is a statistically significant driver of CO2. If it is > 0.05, it doesn't have a proven impact on this specific model.

The coefficient for renewables was 0.000122057, meaning that for every 1% increase in Renewable Energy, Bulgaria CO2 emissions drop by 0.00012 tonnes per capita (so 0.12kg approximately)

For coal, the coefficient was 0.000416218, so approx. 0.41kg per capita of CO2 emissions would be raised every 1% increase in coal usage.

- These are the results for **Spain**:

Regression Statistics								
Multiple R	0.917784136							
R Square	0.842327721							
Adjusted R	0.814503201							
Standard E	0.517288356							
Observations	21							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	3	24.3018913	8.10063	30.27286	4.8249E-07			
Residual	17	4.548983137	0.267587					
Total	20	28.85087443						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.024625486	2.502531208	0.00984	0.992263	-5.2552538	5.304504812	-5.25525384	5.304504812
GDP_Per_Capita	0.000185798	8.64765E-05	2.148541	0.046372	3.3488E-06	0.000368248	3.3488E-06	0.000368248
Total_Renewable_Energy	-0.00031532	0.000183099	-1.72213	0.103188	-0.0007016	7.09844E-05	-0.00070162	7.09844E-05
Coal_Usage	0.000653871	0.000107496	6.082755	1.22E-05	0.00042707	0.000880667	0.00042707	0.000880667

Our R Square here is 0.842327721, meaning that the multiple regression model is highly effective, explaining 84.2% of the variance in Spain's CO2 emissions. Only 15.8% of the changes in CO2 are caused by outside factors not included in this model.

The overall model is highly statistically significant ($p < 0.001$), proving that this combination of variables is a reliable predictor of Spain's emissions.

Coal usage is the most significant driver of CO2 in this model (p is much under 0.001). Every unit increase in the share of coal significantly raises CO2 per capita.

GDP per capita has a positive impact on emissions (p = 0.046). This means that, as Spain's economy grows, emissions tend to rise slightly.

Surprisingly, when controlling GDP and Coal usage simultaneously, the share of Renewable Energy is not a significant predictor of CO2 reductions (p = 0.103).

This suggests that Spain's emissions profile is dominated so heavily by how much Coal they burn, that their renewable additions alone aren't strong enough to predict the CO2 decrease.

- These are the results for **Denmark**:

Regression Statistics								
Multiple R	0.983939195							
R Square	0.968136339							
Adjusted R Square	0.96251334							
Standard Error	0.39159837							
Observations	21							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	3	79.20845097	26.40282	172.1744	6.41707E-13			
Residual	17	2.606937813	0.153349					
Total	20	81.81538878						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	10.09004903	2.610754568	3.864802	0.001243	4.581838376	15.5982597	4.58183838	15.5982597
GDP_Per_Capita	-0.000111132	6.55486E-05	-1.69541	0.108234	-0.000249427	2.7164E-05	-0.0002494	2.7164E-05
Total_Renewable	-0.000115528	0.000155761	-0.7417	0.468393	-0.000444155	0.0002131	-0.0004442	0.0002131
Coal_%	0.000525792	4.64056E-05	11.33036	2.41E-09	0.000427885	0.0006237	0.00042788	0.0006237

Our multiple regression model for Denmark explains a massive 96.8% of the variance in their CO2 emissions. This is an exceptionally strong fit, meaning only 3.2% of the changes in CO2 are due to external factors not captured by this model.

The overall model is highly statistically significant (p < 0.001). This confirms that the chosen variables are overwhelmingly reliable predictors of Denmark's carbon footprint.

Coal usage remains the most highly significant driver of CO₂ emissions ($p < 0.001$). meaning that each unit increase in the share of coal directly and heavily drives up Denmark's CO₂ per capita.

Unlike Spain, GDP per capita is not a predictor of emissions in Denmark ($p = 0.108$). This demonstrates strong economic 'decoupling.' Denmark has reached a level of development where their economic growth is no longer mathematically tied to an increase in carbon emissions.

On the other hand, and similarly to Spain, when controlling for Coal and GDP at the same time, the total share of Renewable Energy is not statistically significant ($p = 0.468$). This reinforces the point that CO₂ reductions in this model are driven primarily by the phasing out of coal, rather than just the addition of renewables. If a country adds wind power but keeps burning coal, emissions will remain high.

6. CLUSTER ANALYSIS

This step is a step-up in difficulty, so the use of excel is not plausible. My move here was to use my knowledge in R and make a short script to calculate the K-Means cluster analytical part.

For this part, I will take a closer look at 10 countries: Spain, Bulgaria, Italy, France, Ireland, Netherlands, China, USA, Germany and Denmark, all from the year 2000 to 2020.

Our first step is to export our data as a .csv file. I selected the data about Country, Year, GDP Per capita, CO2 Per capita and Renewables and exported that to R. I will personally use R 4.3.3 in Visual Studio Code for the coding.

After opening our R script, we need to ensure that the correct statistical libraries are installed. Those are the ones needed for the cluster analysis: cluster and dplyr.

When you look at your dataset, you have three variables for each country: CO2, GDP and Renewables.

To graph all three of those at the same time, you would technically need a 3D graph, but doing so on a flat computer screen can be hard to read.

To fix this, the clusplot function performs a neat mathematical calculation called Principal Component Analysis (PCA).

PCA takes your three variables (CO2, GDP, Renewables) and joins them into two brand-new ones, so they fit perfectly on a standard 2D flat graph.

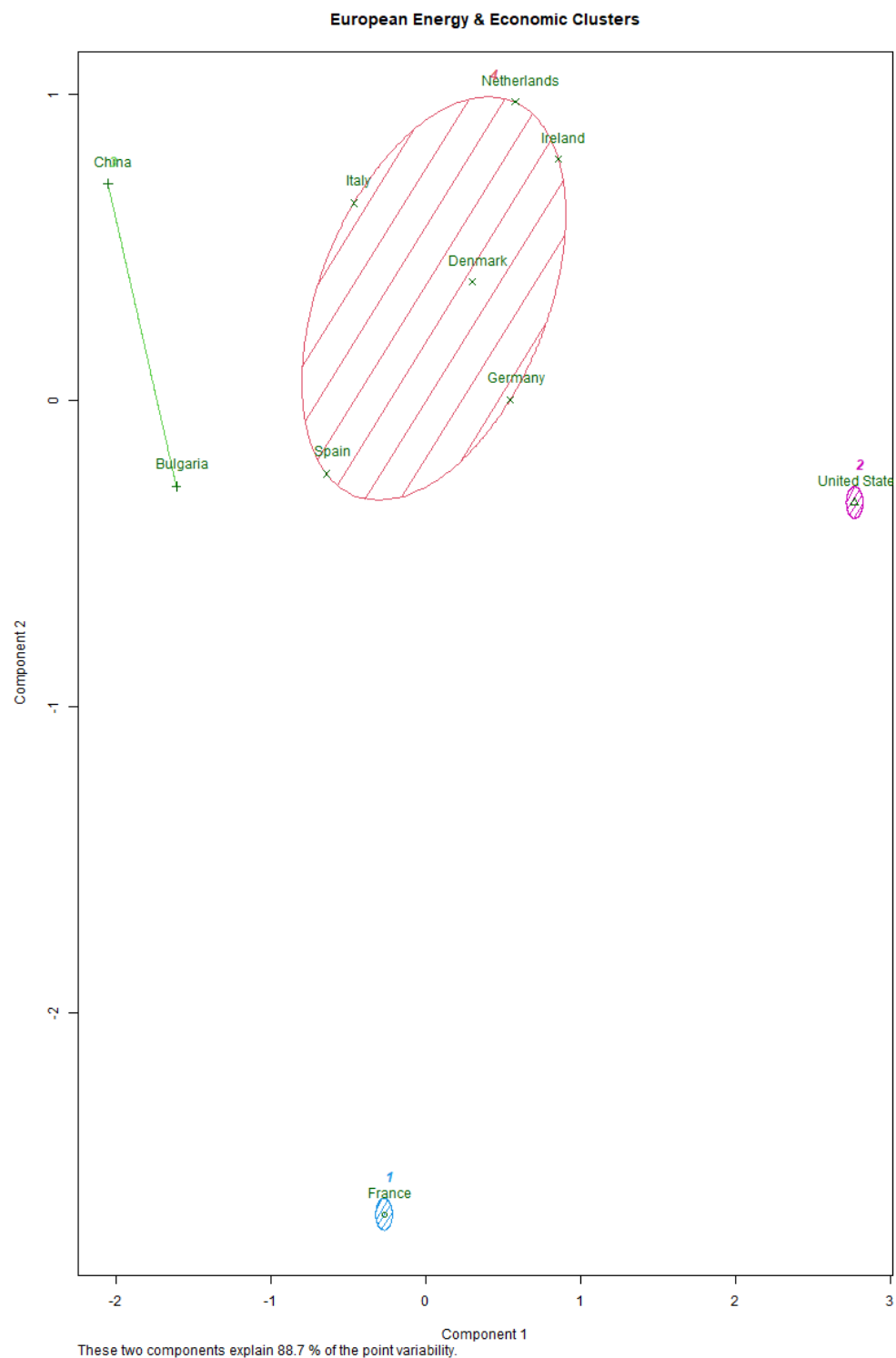
- **Component 1 (The X-Axis):** This is the mathematical combination of your variables that explains the biggest differences between the countries. This is likely heavily influenced by the size of the economy and industry (GDP and CO2).
- **Component 2 (The Y-Axis):** This explains the second biggest difference between the countries. This is capturing the split in how these countries generate their power (the Renewable percentage)

My script for the cluster analysis looked something like this:

```
testfile.R
1 # First we set working directory where we stored our data. This was mine:
2 setwd("C:/Users/diego/Desktop/UniRuse/Statistics")
3
4 # Then we must load the downloaded libraries
5 library(dplyr)
6 library(cluster)
7
8 # Now we can load the data
9 df <- read.csv("euro_data.csv")
10
11 # We group the data to find the 21-year average for each country
12 country_summary <- df %>%
13   group_by(Country) %>%
14   summarise(
15     Mean_CO2 = mean(CO2, na.rm = TRUE),
16     Mean_GDP = mean(GDP, na.rm = TRUE),
17     Mean_Ren = mean(Renewables, na.rm = TRUE)
18   )
19
20 # So that data is converted to a standard data frame (to see the names)
21 country_summary <- as.data.frame(country_summary)
22
23 # Set the row names to the Country names so they appear on our plot
24 rownames(country_summary) <- country_summary$Country
25
26 # Scale the data (Crucial for clustering so GDP doesn't overpower CO2)
27 clustering_vars <- scale(country_summary[, 2:4])
28
29
30 # Run K-Means Clustering (We will group them into 3 distinct clusters)
31 set.seed(123) # Ensures you get the exact same result every time
32 km_res <- kmeans(clustering_vars, centers = 4, nstart = 25)
33
34 # Plot the Clusters
35 clusplot(clustering_vars, km_res$cluster, color=TRUE, shade=TRUE,
36 | | | | labels=2, lines=0, main="European Energy & Economic Clusters")
```

With this code we get the following cluster map, which represents the data for our 10 countries

The plot obtained with it is this one:



We can see we obtained four separate clusters, which align with real-world differences in energy policies or economic structuring. This is thanks to dplyr, which helps with cleaning, filtering, and organizing data and then cluster which is the specific library.

- **Cluster 1: The core European model.**

Nations: Netherlands, Ireland, Germany, Denmark and Spain.

The algorithm grouped most of the Western and Northern European nations into one central cluster. Despite their individual differences, these countries share a unified profile: high GDP per capita, paired with active renewable energy transitions and a steady decline in carbon footprints. They represent the standard "European" approach to modern energy economics.

- **Cluster 2: The high consumption superpowers**

Nations: United States

The USA exists in an isolated cluster on the extreme far right of Component 1. This mathematical isolation is driven by its unique profile as a global superpower: a massive GDP per capita combined with historically high baseload energy consumption and per capita CO2 emissions that vastly exceed the European average.

- **Cluster 3: The Nuclear Outlier**

Nation: France.

France is completely separated from its Western European neighbors, sitting at the extreme bottom of Component 2. This is because France's power grid is overwhelmingly dominated by nuclear energy. Therefore, it achieves the high GDP of a Western European nation but with a drastically lower CO2 per capita baseline.

- **Cluster 4: High carbon emerging countries**

Nations: China and Bulgaria

Grouped together on the far left of Component 1, these two nations share a distinct mathematical similarity in this specific dataset. Compared to the USA or Western Europe, both possess a lower average GDP per capita over the 2000-2020 timeframe, but maintain a high reliance on coal and fossil fuels for industrial growth. This results in a high carbon profile linking the Eastern European energy baseline with the massive industrial expansion of China.

- **Conclusion:** The introduction of global superpowers like the USA and China into the cluster analysis validates the existence of a distinct "European Energy Transition" profile. While countries like Germany and Spain may differ internally, on a global scale, they move together. Furthermore, the model did an amazing job analyzing the data, separating anomalies such as France's nuclear grid and the USA's high-consumption economy.

7. TIME SERIES ANALYSIS (ARIMA FORECAST)

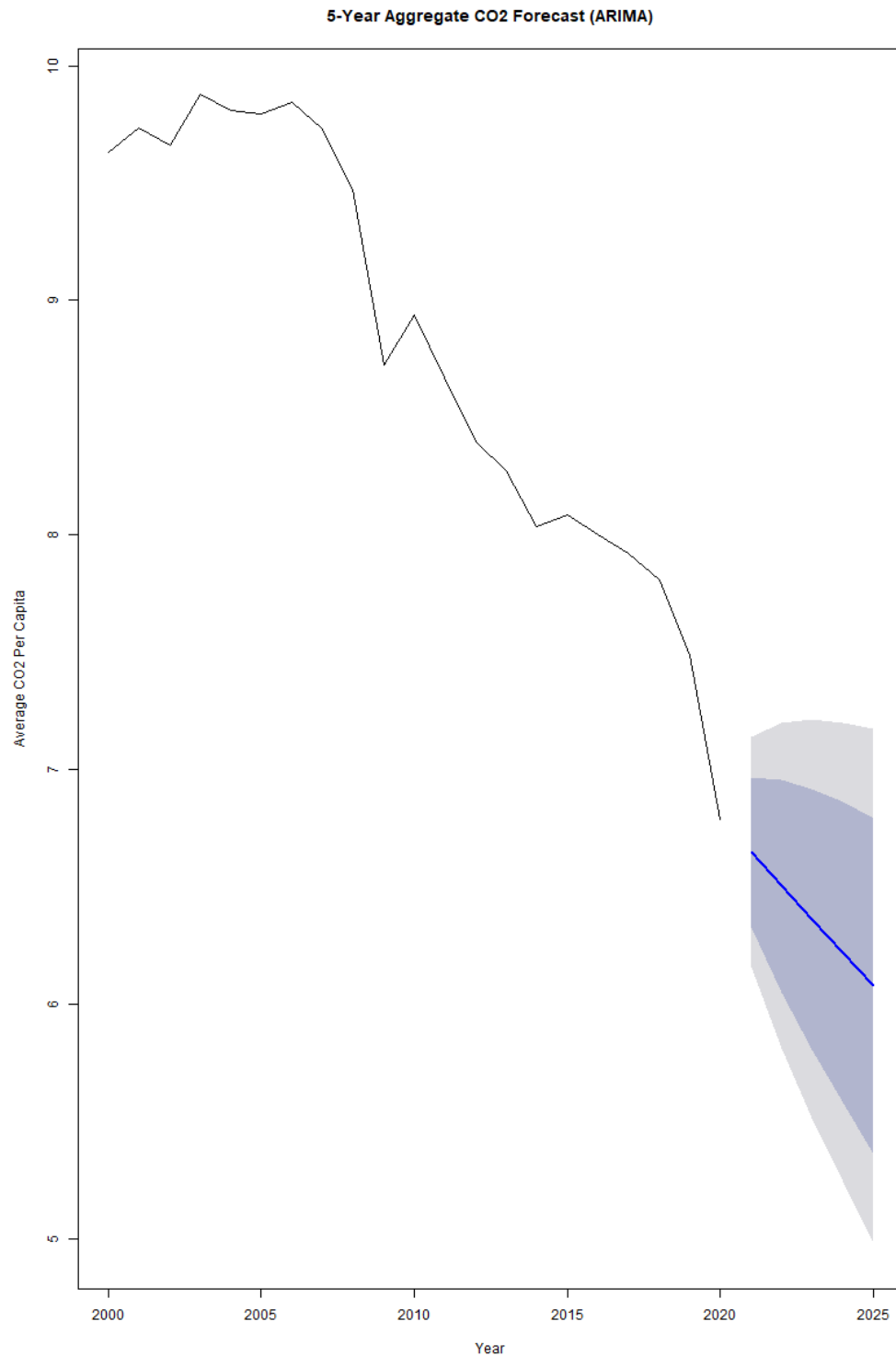
For this analysis we need two new libraries: forecast (for time series prediction), and then tseries (to handle time-based sequential data). ARIMA means **Autoregressive Integrated Moving Average**. If we separate it into parts, it basically is:

- **Autoregressive:** It looks at previous years and assumes the CO2 from 2020 is highly influenced by the one of 2019.
- **Integrated:** Instead of looking exactly at the CO2 number, it looks at the difference between years to find a stable trend.
- **Moving Average:** It corrects mathematical errors the prediction may have, correcting the peaks that past years might have had. That makes the slopes look cleaner and smoother.

In the code, the `auto.arima()` algorithm was used to identify the optimal parameters (p, d, q) based on autocorrelation and historical data, generating a 5-year forecast (2021–2025). Here is the code I used to make the ARIMA prediction:

```
data.r
1 # We first load the needed libraries
2 library(dplyr)
3 library(forecast)
4
5 # Global CO2 average per year (2000-2020)
6 yearly_co2 <- df %>%
7   group_by(Year) %>%
8   summarise(Avg_CO2 = mean(CO2, na.rm = TRUE))
9
10 # Converting those data into a time series object (This is needed for ARIMA)
11 co2_ts <- ts(yearly_co2$Avg_CO2, start = 2000, end = 2020, frequency = 1)
12
13 # Create the ARIMA model automatically (R will find the best mathematical formula)
14 arima_model <- auto.arima(co2_ts)
15
16 # Predict the next 5 years of CO2 emissions using the ARIMA model
17 co2_forecast <- forecast(arima_model, h = 5)
18
19 # Plot the forecast
20 plot(co2_forecast,
21       main="5-Year Aggregate CO2 Forecast (ARIMA)",
22       ylab="Average CO2 Per Capita",
23       xlab="Year",
24       col="black",      # Actual data line color
25       fcol="blue")      # Prediction line color
```

The resulting plot is this one:



The gray areas seen in the graphic are the confidence intervals (usually 95%). This means that, as we do not know the exact number of the future CO2 per capita, we can at least be sure it will be in that interval 95 out of 100 times. We can notice too that the gray zone grows bigger as years advance. This is because predicting 2025 is more unsure than 2021 (with data until 2020).

8. CONCLUSIONS

This statistical project evaluated the dynamics between economic growth (GDP per capita), renewable energy adoption, and CO2 emissions per capita across ten diverse global economies from the years 2000 to 2020.

I can conclude that the data supports the hypothesis that transitioning to renewable energy is vital for reducing carbon footprints. Nevertheless, the statistical models prove that adding renewable capacity is only half the equation. The strongest path to reducing per capita emissions remains the phase-out of coal and the decoupling of economic growth from carbon energy sources.